



# CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE(Main)

FULL TEST

20-12-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : TEAS, TOAS, TAAS, TNAS, I(A) &amp; LIVE-I(A)] (SCORE-I)

Time : 3 Hours

## PAPER-2

Maximum Marks : 300

READ THE INSTRUCTIONS CAREFULLY

### Important Instructions :

1. Immediately fill in the form number on this page of the Test Booklet with Blue/Black Ball Point Pen. Use of pencil is strictly prohibited.
2. The candidates should not write their Form Number anywhere else (except in the specified space) on the Test Booklet/Answer Sheet.
3. The Test Booklet consists of **90** questions.
4. There are **three** parts in the question paper 1,2,3 consisting of **Physics, Chemistry and Mathematics** having **30 questions** in each subject and each subject having **Two sections**.
  - (i) Section-I contains 20 multiple choice questions with **only one correct** option.  
**Marking scheme** : +4 for correct answer, 0 if not attempted and -1 in all other cases.
  - (ii) Section-II contains 10 **Numerical Value Type** questions. Attempt any 5 questions. First 5 attempted questions will be considered for marking.  
**Marking scheme** : +4 for correct answer, 0 if not attempted and -1 in all other cases.
5. Use **Blue/Black Ball Point Pen only** for writing particulars/markings responses on **Side-1** and **Side-2** of the Answer Sheet. **Use of pencil is strictly prohibited.**
6. No candidate is allowed to carry any textual material, printed or written, bits of papers, mobile phone any electronic device etc, except the Identity Card inside the examination hall/room.
7. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
8. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Room/Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
9. **Do not fold or make any stray marks on the Answer Sheet.**
10. Take  $g = 10 \text{ m/s}^2$  unless otherwise stated.

DO NOT BREAK THESE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

Name of the Candidate (in Capitals)\_\_\_\_\_

Form Number : in figures\_\_\_\_\_

: in words\_\_\_\_\_

Centre of Examination (in Capitals) : \_\_\_\_\_

Candidate's Signature : \_\_\_\_\_ Invigilator's Signature : \_\_\_\_\_

## Your Target is to secure Good Rank in JEE(Main) 2023

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PART-1 : PHYSICS

SECTION-I : (Maximum Marks: 80)

This section contains **20 questions**. Each question has 4 options for correct answer. Multiple-Choice Questions (MCQs) **Only one option is correct**. For each question, marks will be awarded as follows:

**Full Marks** : +4 If correct answer is selected.

**Zero Marks** : 0 If none of the option is selected.

**Negative Marks** : -1 If wrong option is selected.

1. Regarding x-rays, choose the correct statement(s) :

I. Moseley's experiment helped to clear several discrepancies in the arrangement of elements in periodic table.

II. Using wavelength of continuous x-ray, atomic number of target metal can be found.

III. In Moseley's law  $\sqrt{\nu} = a(Z - b)$ ,  $b$  is independent but  $a$  depends on the material.

IV. When passed through a gas chamber x-rays ionize the molecules of the gas.

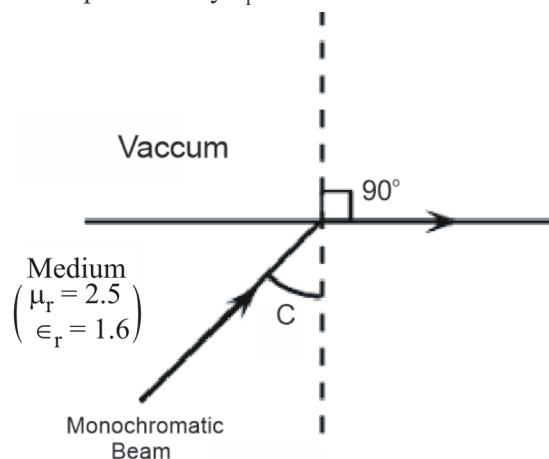
(A) I, III, IV

(B) I, IV

(C) I, II

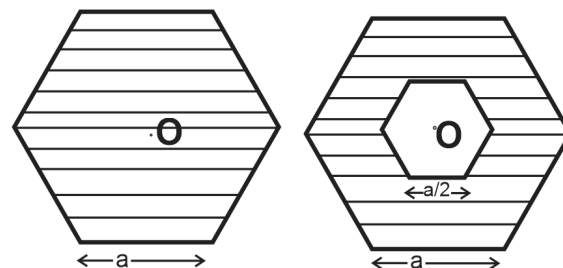
(D) only I

2. A monochromatic light beam is incident from an optically denser medium to vacuum at critical angle. Find its value (in degrees) if relative permeability of medium  $\mu_r = 2.5$  and relative permittivity  $\epsilon_r = 1.6$



(A)  $37^\circ$  (B)  $45^\circ$  (C)  $30^\circ$  (D)  $53^\circ$

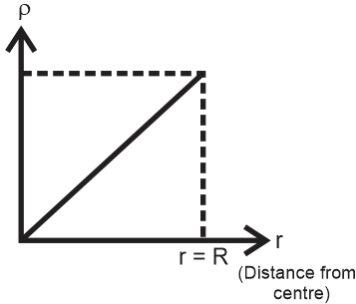
- 3.



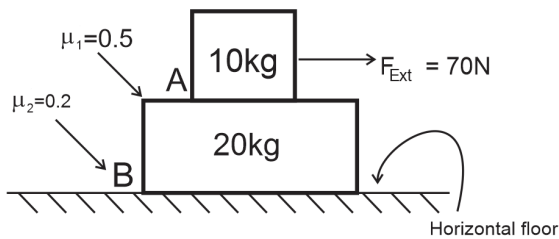
Moment of inertia of a uniform hexagonal plate about an axis through its centre 'O' and perpendicular to its plane is  $I_0$ . If a small hexagon of side length  $\frac{a}{2}$  is removed as shown, then moment of inertia of the remaining part about the same axis through 'O' is :-

(A)  $\frac{5I_0}{6}$  (B)  $\frac{3I_0}{4}$   
(C)  $\frac{15I_0}{16}$  (D)  $\frac{7I_0}{8}$

4. Charge  $4Q$  is distributed in a sphere of radius  $R$ , with variable charge density that varies with distance from centre as shown in figure. Two point charges A and B ( $-Q$  each) are placed at diametrically opposite points, at equal distance  $x$ , from the centre of sphere. Individually, both A and B do not experience any electrostatic force. Then

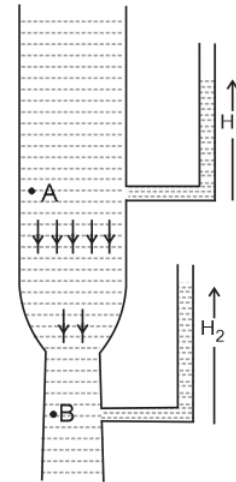


- (A)  $R = 2(2)^{1/4}x$   
 (B)  $R = 8^{1/4}x$   
 (C)  $R = 4x$   
 (D)  $R = 2x$
5. Find  $|\vec{a}_A - \vec{a}_B|$ . Symbols have usual meaning. Take  $g = 10 \text{ m/s}^2$

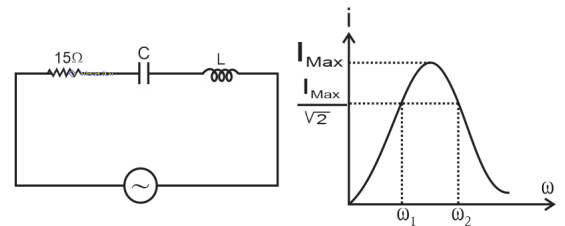


- (A)  $2 \text{ m/s}^2$   
 (B)  $1 \text{ m/s}^2$   
 (C)  $3 \text{ m/s}^2$   
 (D) None

6. Water flows through a vertical tube of variable cross section. The area of cross section at A is  $600 \text{ cm}^2$  and at B is  $150 \text{ cm}^2$ . Vertical distance between points A and B is  $H = 80 \text{ cm}$ . Also,  $H_2 - H_1 = 5 \text{ cm}$ . Find volume flow rate of water through the tube. Assume ideal fluid behaviour. Take  $g = 10 \text{ m/s}^2$ ,  $\rho = 1000 \text{ kg/m}^3$ ,  $P_{\text{atm}} = 10^5 \text{ N/m}^2$ .

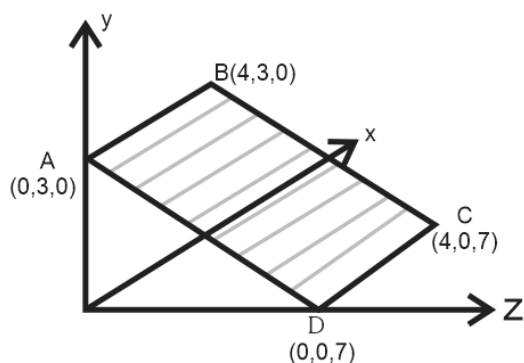


- (A)  $60000 \text{ cm}^3/\text{sec}$  (B)  $40000 \text{ cm}^3/\text{sec}$   
 (C)  $50000 \text{ cm}^3/\text{sec}$  (D)  $80000 \text{ cm}^3/\text{sec}$
7. In a series LCR circuit the frequencies at which the current amplitude becomes  $\frac{I_{\text{max}}}{\sqrt{2}}$  are  $\omega_1 = 296 \text{ rad/sec}$  and  $\omega_2 = 316 \text{ rad/sec}$ . The value of resistance is  $R = 15\Omega$ . Inductance (in mH) of inductor connected in the circuit is



- (A) 750 (B) 600  
 (C) 250 (D) 200

8. A beam of plane polarized, monochromatic light wave is incident normally from vacuum on a sapphire slab (non-magnetic,  $n = 7/4$ ). 37 % of light gets reflected in vacuum. Find the amplitude of Electric field for the transmitted wave if the amplitude of Electric field corresponding to incident light is 72 V/m.
- (A) 43.2 V/m  
(B) 41.14 V/m  
(C) 12.34 V/m  
(D) 48.6 V/m
9. A monochromatic parallel light beam of intensity  $200 \text{ Wm}^{-2}$  along +ve z-axis is incident on surface ABCD whose coordinates (in meters) are mentioned in figure. Find the number of photons crossing the surface ABCD in time interval of 5 seconds. Take  $\lambda = 500 \text{ nm}$  and  $h = 6.63 \times 10^{-34} \text{ J-s}$



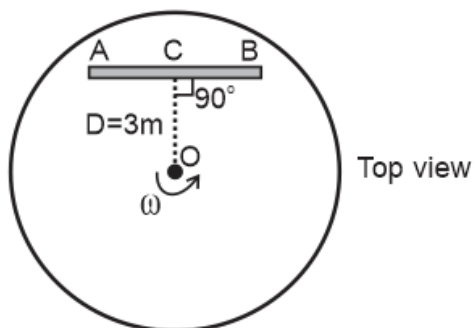
- (A)  $9.82 \times 10^{22}$   
(B)  $3.01 \times 10^{22}$   
(C)  $5.88 \times 10^{21}$   
(D)  $8.68 \times 10^{21}$

10. A spy satellite is orbiting at an altitude of 200 kms around Earth in a circular orbit. It is equipped with a telescope of aperture diameter 2.4m facing towards Earth. Find the minimum distance (approx) between two neighbouring small monochromatic light sources of wavelength  $\lambda = 550 \text{ nm}$  on Earth, so that the telescope can form their distinct images.
- (A) 5.6 cms (B) 4.54 cms  
(C) 7.6 cms (D) 12 cms
11. Find percentage change in the weight of a block when it is taken to an altitude of 16 km above the surface of Earth. Radius of Earth = 6400 km
- (A) 0.5 % (B) 0.4 %  
(C) 0.8 % (D) 1 %
12. Pick the correct statement(s) from given four statement(s)
- (I) In a PN junction diode, during forward biasing drift current exceeds diffusion current in magnitude.  
(II) In normal operation, photodiode operates in forward bias condition.  
(III) Semiconductors with band gap close to 1.5 eV are ideal materials for solar cell fabrication.  
(IV) In normal operation, LEDs operates in forward bias condition.
- (A) (I), (II) and (III)  
(B) (I), (II) and (IV)  
(C) (II), (III) and (IV)  
(D) (III) and (IV)

13. 250 gms of an ideal diatomic gas sample is at a pressure of  $4 \times 10^4 \text{ N/m}^2$  in a closed beaker. Density of gas is  $6 \text{ kg/m}^3$ . Total internal energy of gas sample is

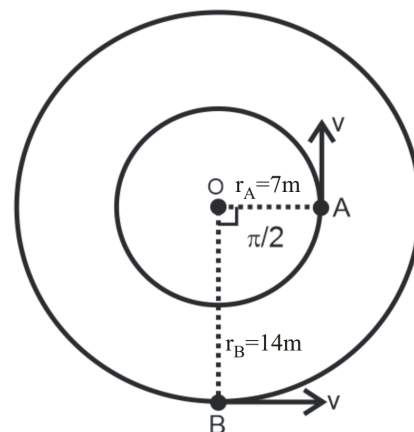
(A) 2 kJ  
(B)  $\frac{15}{6}$  kJ  
(C)  $\frac{25}{6}$  kJ  
(D) 15 kJ

14. A uniform rod AB of length  $2\sqrt{3} \text{ m}$  and mass 5 kg is placed on a rough platform rotating in horizontal plane with constant angular speed  $\omega = 4 \text{ rad/sec}$ , about an axis through centre of platform and perpendicular to its plane. The centre of rod (C) is at a distance  $D = 3 \text{ m}$  from the centre of platform. Find total kinetic energy of rod if it does not slip on the platform.



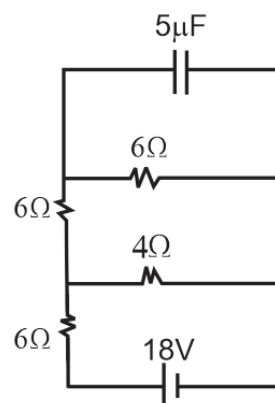
(A) 40 J  
(B) 200 J  
(C) 400 J  
(D) 500 J

15. Two particles A and B are moving along two concentric co-axial circles of radius  $r_A = 7 \text{ m}$  and  $r_B = 14 \text{ m}$  with equal linear speed  $v = 5 \text{ m/sec}$ . At  $t = 0$ , their positions are as shown in the figure. After how many seconds, centre O, particle A and particle B will be collinear for the first time (Take  $\pi = \frac{22}{7}$ ).



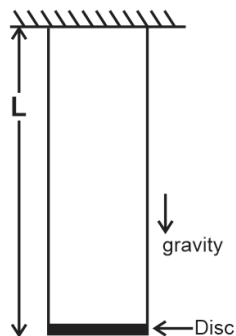
(A)  $\frac{22}{5}$  sec  
(B)  $\frac{22}{15}$  sec  
(C)  $\frac{44}{15}$  sec  
(D)  $\frac{66}{5}$  sec

16. Find charge on capacitor in the given circuit in steady state.



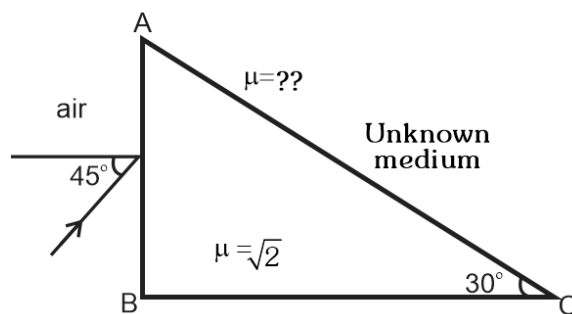
(A)  $15 \mu\text{C}$   
(B)  $10 \mu\text{C}$   
(C)  $12 \mu\text{C}$   
(D)  $20 \mu\text{C}$

17. A uniform rod of mass  $M$ , length  $L$ , cross-section area  $A$  has Young's modulus of elasticity  $Y$ . It is suspended vertically from a fixed roof as shown. A disc of negligible thickness and mass same as that of rod, is welded to the rod at its bottom. Find total extension in the length of rod.



- (A)  $\frac{3MgL}{2AY}$  (B)  $\frac{2MgL}{AY}$   
 (C)  $\frac{MgL}{2AY}$  (D)  $\frac{MgL}{AY}$
18. Consider three quantities  $x = \frac{E}{B}$ ,  $y = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$  and  $z = \frac{\ell}{CR}$ . Here,  $\ell$  is the length of wire,  $C$  represents capacitance and  $R$  is resistance. All other symbols have standard meaning. From the given below statements, choose correct option(s).
- (I)  $x, y$  have the same dimension  
 (II)  $y, z$  have the same dimension  
 (III)  $z, x$  have the same dimension  
 (IV) None of the three have same dimensions
- (A) (I), (II) and (III)  
 (B) (I) and (II) only  
 (C) (II) and (III) only  
 (D) (IV) only

19. A light ray is incident from air ( $\mu = 1$ ) to a transparent right angled prism  $ABC$  ( $\mu_{\text{prism}} = \sqrt{2}$ ) as shown in the figure. The ray emerges out from face  $AC$  of the prism to a medium of unknown refractive index such that the ray suffers no net deviation due to prism. Find refractive index of unknown medium.



- (A)  $2\sqrt{3}$   
 (B)  $\sqrt{3} + 1$   
 (C)  $\sqrt{6}$   
 (D)  $\frac{3\sqrt{3}}{2}$
20. A ball thrown upwards from the top of a tower with initial speed  $20 \text{ m/s}$  reaches base of the tower in next  $6 \text{ sec}$ . Maximum height above ground level reached by ball is (Take  $g = 10 \text{ m/s}^2$ ).
- (A)  $80 \text{ m}$   
 (B)  $60 \text{ m}$   
 (C)  $20 \text{ m}$   
 (D)  $120 \text{ m}$

SECTION-II : (Maximum Marks: 20)

This section contains 10 questions. Candidates have to attempt any 5 questions out of 10. If more than 5 questions are attempted, then only first 5 attempted questions will be evaluated.

The answer to each question is an Integer Value Type questions.

For each question, enter the correct integer value (In case of non-integer value, the answer should be rounded off to the nearest Integer).

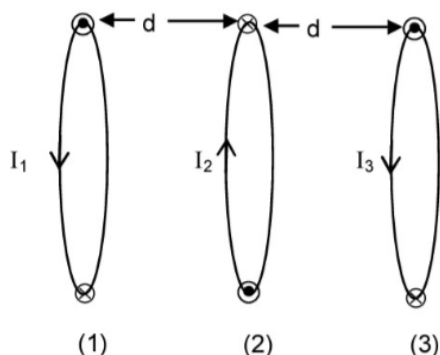
Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If correct answer is selected.

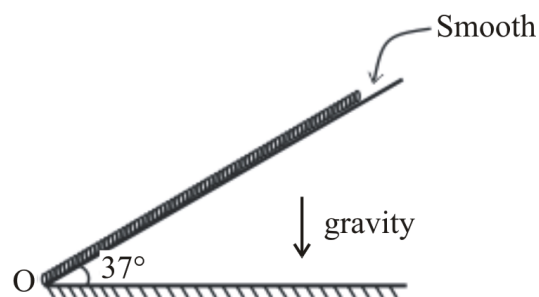
Zero Marks : 0 If none of the option is selected.

Negative Marks : -1 If wrong option is selected.

1. Consider 3 identical conducting loops kept parallel to each other such that their axis coincides. Loop (1), loop (2) and loop (3) have self inductance equal to  $L$  each. Current in loop (1) is  $I_1$ , in loop (2) is  $I_2$  and in loop (3) it is  $I_3$ . Rate of change of current in them at certain instant is  $\frac{dI_1}{dt}$ ,  $\frac{dI_2}{dt}$ ,  $\frac{dI_3}{dt}$  respectively. Their mutual inductance is  $M_{ij}$  (i.e. for 1 and 2 its  $M_{12}$ ). Current in loop (1) and (3) is in same sense and in (2) it is in opposite sense. If  $\frac{dI_1}{dt} = \frac{dI_2}{dt} = \frac{dI_3}{dt}$  and emf induced in loop (2) is zero, find  $\frac{L}{M_{12}}$ .

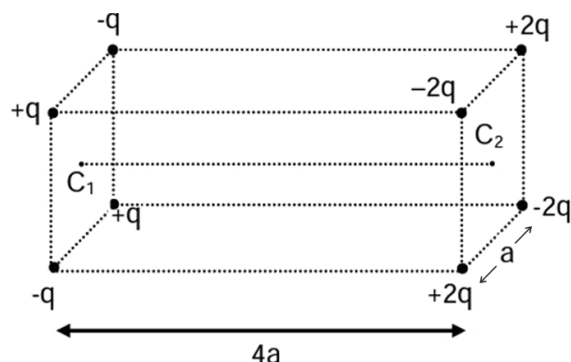


2. A wooden sphere of density  $750 \text{ kg/m}^3$  and radius  $1.5 \text{ cm}$ , rises steadily through a viscous liquid of density  $\rho_L = 1750 \text{ kg/m}^3$  at a rate of  $80 \text{ cm/sec}$ . If the coefficient of viscosity of liquid in poise is  $\eta$ , find  $100\eta$ . Take  $g = 10 \text{ m/s}^2$ .
3. A uniform chain of length  $L = 9 \text{ m}$  is released from rest on a fixed, smooth incline as shown in figure. Initially the lower end of the chain is at the lower edge 'O' of incline. Determine the speed  $v$  (in  $\text{m/sec}$ ) (approximated to nearest integer) of the chain when 40 % of the chain has gone below point 'O' and hanging vertically. (Neglect edge effect). Take  $g = 10 \text{ m/s}^2$ .

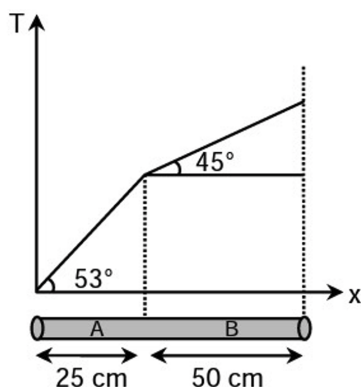


4. Two radioactive nuclei A and B decay into C with decay constant  $\lambda_A = \lambda$  and  $\lambda_B = 2\lambda$  respectively. Initially, number of active nuclei of A is equal that of B. Initial rate of production of C is  $x$  times the rate of production of C after time  $t = \frac{\ln 2}{\lambda}$ . Find  $x$ .

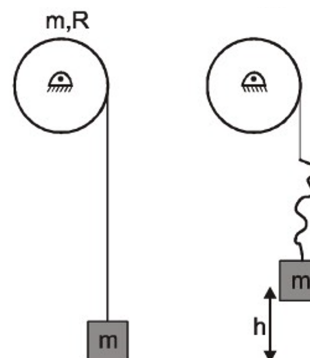
5. Two parallel square plates each of side length 'a' are arranged such that line joining centre of both the plates is perpendicular to planes of both plates. Eight fixed charges are placed at corners of plate as shown in figure. A charged particle of charge +q and mass m is projected with initial velocity  $V_0$  from centre  $C_1$  of one square plate towards centre  $C_2$  of other. The length of line  $C_1C_2$  is  $4a$ . The speed of charged particle at mid-point of line  $C_1C_2$  is  $\frac{4V_0}{x}$ . Find the value of x ? (Neglect effect of gravity).



6. Two conductors A and B each of cross sectional area  $5\text{cm}^2$  are connected in series. Variation of temperature (in  $^\circ\text{C}$ ) along the length (in cm) is as shown in the figure. If thermal conductivity of A is  $120\text{ J/m-sec-}^\circ\text{C}$ , the thermal conductivity of B (in SI units) is :



7. Potential energy of a mechanical system as a function of time t is given by  $E(t) = K^2 e^{-\alpha t}$ , where  $\alpha = 0.2\text{s}^{-1}$ . The measurement of K has an error of 1.75% and measurement of time t has an error of 1.50%. Find the percentage error in the value of E(t) at  $t = 5\text{ sec}$ .
8. Calculate the binding energy of an alpha particle (in MeV) to nearest integer from the given data : Take  $1\text{u} = 931\text{ MeV}/c^2$   
 Mass of  $^1_1\text{H}$  atom =  $1.007813\text{ u}$   
 Mass of neutron =  $1.008665\text{ u}$   
 Mass of He atom =  $4.00260\text{ u}$
9. A solid cylindrical pulley of mass m and radius  $R = 10\text{ cm}$  is hinged about its horizontal axis of symmetry. A light string is wrapped around it, and a small block of mass 'm' is suspended from the string. Now the block is lifted vertically by a distance  $h = 1.8\text{ m}$  and released. Just after the string becomes taut again, find the angular velocity of the cylinder in rad/s. (Take  $g = 10\text{ m/sec}^2$ )



10. What is the short-circuit current (approximated to nearest integer) delivered by a  $10\text{ cm}$  by  $10\text{ cm}$  photocell (with 100% quantum efficiency) illuminated by monochromatic light of  $400\text{ nm}$  wavelength with a power density of  $1000\text{ W/m}^2$ . Assume normal incidence. Take  $h = 6.62 \times 10^{-34}\text{ Js}$ .

SECTION-I : (Maximum Marks: 80)

This section contains **20 questions**. Each question has 4 options for correct answer. Multiple-Choice Questions (MCQs) **Only one option is correct**. For each question, marks will be awarded as follows:

**Full Marks** : +4 If correct answer is selected.

**Zero Marks** : 0 If none of the option is selected.

**Negative Marks** : -1 If wrong option is selected.

1. Which of the following statement(s) is/are correct ?

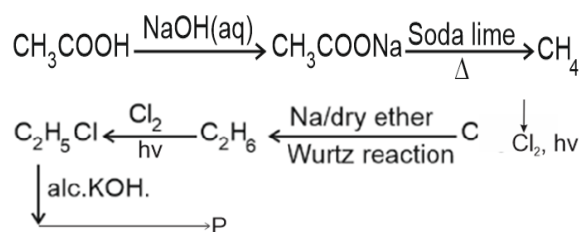
- (A)  $[\text{Co}(\text{NH}_3)_4(\text{H}_2\text{O})_2]\text{Cl}_2$  has geometrical isomers  
(B)  $[\text{Co}(\text{C}_2\text{O}_4)_3]^{3-}$  is paramagnetic.  
(C)  $[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]$  is paramagnetic.  
(D)  $[\text{Ni}(\text{H}_2\text{O})_6]\text{Cl}_2$  has geometrical isomers.

2. Which of the following statement(s) is/are incorrect?

- (I) For hydrogen gas ( $P = 200 \text{ atm}$ ,  $T = 273 \text{ K}$ ) compressibility factor  $\neq 1$   
(II) For hydrogen gas ( $P \sim 0$ ,  $T = 273 \text{ K}$ )  $P(V - nb) = nRT$   
(III) Protein is a natural polymer and has glycosidic linkage  
(IV) Cellulose is a synthetic polymer

- (A) I only  
(B) I and II only  
(C) III and IV only  
(D) II, III and IV only

3. Consider the following reactions



P is used to synthesize benzene through the reagents given in the following sequence :

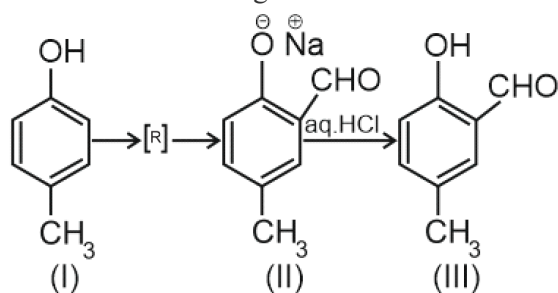
- (A) (i)  $\text{Br}_2$  (ii) alc. KOH (iii) Red hot iron tube,  $873 \text{ K}$   
(B) (i)  $\text{Br}_2$  (ii) alc. KOH (iii)  $\text{NaNH}_2$  (iv) Red hot iron tube,  $873 \text{ K}$   
(C) (i)  $\text{Br}_2$  (ii)  $\text{NaNH}_2$  (iii) Conc.  $\text{HNO}_3$   
(D) (i)  $\text{Br}_2$  (ii) Red hot iron tube,  $873 \text{ K}$  (iii) alc. KOH (iv)  $\text{NaNH}_2$

4. Consider a reaction  $a\text{A} + b\text{B} \rightarrow c\text{C} + \text{D}$ . When concentration of A as well as B is doubled, the rate of the reaction increases by 8 times. However, when concentration of B is doubled keeping the concentration of A fixed, the rate is doubled. The partial order of reaction with respect to B is

- (A) 1  
(B) 2  
(C) 3  
(D) Zero

5. In which one of the following equilibria,  $K_p = K_c$ ?
- (A)  $\text{NO}_2(\text{g}) + \text{SO}_2(\text{g}) \rightleftharpoons \text{NO}(\text{g}) + \text{SO}_3(\text{g})$
- (B)  $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$
- (C)  $2\text{C}(\text{s}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{CO}(\text{g})$
- (D)  $\text{CO}_2(\text{g}) \rightleftharpoons \text{CO}_2(\text{aq})$
6. Which of the following statements is incorrect?
- (A) The reaction  $\text{O}_2^- \rightarrow \text{O}_2 + \text{O}_2^{2-}$  is a disproportionation reaction
- (B) The reaction  $\text{CrO}_4^{2-} + \text{H}^+ \rightarrow$  gives a dimeric bridged tetrahedral metal ion
- (C) The reaction  $\text{MnO}_4^- + \text{NO}_2^- + \text{H}^+ \rightarrow$  gives multiple products. One of these products containing nitrogen has linear structure
- (D) The reaction  $\text{NO}_3^- + \text{H}_2\text{SO}_4 + \text{Fe}^{2+} \rightarrow$  gives a brown ring in the test-tube
7. 75.2 g of phenol is dissolved in 1 kg of a non polar solvent. If 75% dimerization of phenol in this solvent leads to 7 K of depression in freezing point. Then Cryoscopic Constant ( $K_f$ ) of the unknown solvent is :
- (A)  $24 \text{ K kg mol}^{-1}$
- (B)  $14 \text{ K kg mol}^{-1}$
- (C)  $2.4 \text{ K kg mol}^{-1}$
- (D)  $1.4 \text{ K kg mol}^{-1}$
8. Which of the following statement(s) is/are correct?
- (i) Even a low concentration of sulphur causes respiratory diseases e.g. bronchitis, in human beings.
- (ii) When nitric oxide reacts with ozone in the stratosphere, the reaction is as follows :  
 $\text{NO} + \text{O}_3 \rightarrow \text{N}_2 + 2\text{O}_2$
- (iii) Besides carbon dioxide, other greenhouse gases are methane, water vapour, nitrous oxide, CFCs and ozone.
- (A) (i) only
- (B) (i) and (ii) only
- (C) (ii) and (iii) only
- (D) (i) and (iii) only
9. Which of the following statements is correct?
- (A)  $\text{CN}^-$  is an electrophile
- (B)  $\text{C}_6\text{H}_5\text{CHO}$  gives precipitate with 2, 4-dinitrophenylhydrazine and undergoes cyanohydrin formation
- (C)  $\text{CN}^-$  gives precipitate with  $\text{AgNO}_3$
- (D)  $\text{CH}_3\text{C} \equiv \text{CH}$  does not give precipitate with  $\text{AgNO}_3$

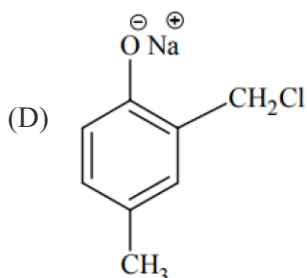
10. Consider the following reaction :



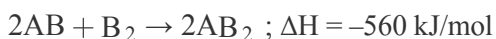
Which of the following statements is correct?

- (A) Mixture of caustic soda and chloroform is used in the above reaction  
 (B) The electrophile in this reaction is  $^+\text{CHCl}_2$   
 (C) The nucleophile in this reaction is  $:\text{CCl}_2$

The structure of the intermediate [R] is



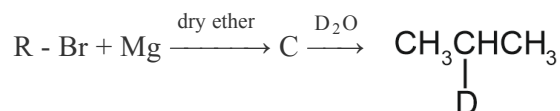
11. The decrease in enthalpy of the following reaction between gaseous reactants at 500 K is :



When two moles of AB and one mole of  $\text{B}_2$  react in a closed and rigid vessel of volume 1 litre, they completely form two moles of  $\text{AB}_2$ , a linear triatomic gas. Due to this reaction the pressure in the vessel changes from 70 atm to 40 atm, the absolute value of change in internal energy due to reaction at 500 K is (1 L.atm = 0.1 kJ) :

- (A) 557 J                                      (B) 557 kJ  
 (C) 5.57 kJ                                      (D) 55.7 kJ

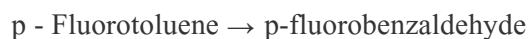
12. Consider the following reaction :



RBr is

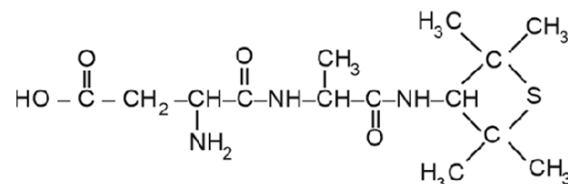
- (A) Bromobutane  
 (B) 2-Bromopropane  
 (C) Bromopropane  
 (D) 2-Bromobutane

13. Which reagent can bring about the following reaction :



- (A) DIBAL-H  
 (B) 1.  $\text{O}_3$  2.  $\text{H}_2\text{O-Zn dust}$   
 (C) 1.  $\text{CrO}_2\text{Cl}_2/\text{CS}_2$  2.  $\text{H}_3\text{O}^+$   
 (D)  $\text{C}_5\text{H}_5\text{NH}^+\text{CrO}_3\text{Cl}^-$  (PCC)

14. The following molecular structure belongs to :



- (A) Anti-depressant  
 (B) Anti-histamine  
 (C) Antiseptic  
 (D) Artificial sweetener

15. Which of the following is correct increasing order of  $E^0$  ( $M^{2+} / M$ ), where M is a 3d transition metal?

(A)  $Mn < Cr < Fe < Cu$   
 (B)  $Cr < Mn < Fe < Cu$   
 (C)  $Mn < Cr < Cu < Fe$   
 (D)  $Cr < Mn < Cu < Fe$

16. Match the following :

| Column - I                                                                                                                                                     | Column - II      |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
|  + $CH_3COCl$<br>Acetylchloride $\xrightarrow[\Delta]{\text{Anhyd. } AlCl_3}$ | (P) Benzophenone |
|  + Benzoyl chloride $\xrightarrow[\Delta]{\text{Anhyd. } AlCl_3}$             | (Q) Toluene      |
|  + $CH_3Cl$ $\xrightarrow{\text{Anhyd. } AlCl_3}$                            | (R) Acetophenone |
|  + $C_2H_5Cl$ $\xrightarrow{\text{Anhyd. } AlCl_3}$                         | (S) Ethylbenzene |

(A)  $A \rightarrow R, B \rightarrow Q, C \rightarrow S, D \rightarrow R$   
 (B)  $A \rightarrow R, B \rightarrow Q, C \rightarrow R, D \rightarrow S$   
 (C)  $A \rightarrow R, B \rightarrow P, C \rightarrow Q, D \rightarrow S$   
 (D)  $A \rightarrow R, B \rightarrow R, C \rightarrow Q, D \rightarrow S$

17. A saturated solution of a sparingly soluble salt XY is taken.  $K_{sp}$  of XY is  $12 \times 10^{-14}$ . If  $1 \times 10^{-7}$  mole of XZ are added to 1 litre of this solution, the conductivity of  $X^+$  ion in this solution is

Given,

| Ion   | Limiting molar conductivity<br>( $S\ m^2\ mol^{-1}$ ) |
|-------|-------------------------------------------------------|
| $X^+$ | $6 \times 10^{-3}$                                    |
| $Y^-$ | $8 \times 10^{-3}$                                    |
| $Z^-$ | $7 \times 10^{-3}$                                    |

(A)  $24 \times 10^{-7}\ S\ m^{-1}$   
 (B)  $24 \times 10^{-8}\ S\ m^{-1}$   
 (C)  $24 \times 10^{-9}\ S\ m^{-1}$   
 (D)  $24 \times 10^{-10}\ S\ m^{-1}$

18. The number of structural isomers for  $C_6H_{14}$  is

(A) 2 (B) 3  
 (C) 4 (D) 5

19. Which of the following statements is correct?

(A) All chiral molecules have chiral centres.  
 (B) Alkali metals dissolves in liquid ammonia to give blue solution because alkali metals in liquid ammonia give solvated blue species of the type  $[M(NH_3)_a]^+$  (M = alkali metals).  
 (C) Sucrose gives a reddish-brown precipitate with Fehling's solution.  
 (D) Band gap in germanium is small compared to insulators

20. The presence of a fluid system is a linear function of volume for a process :  $P + a + bV$  [a & b are constants]. Internal energy of the fluid depends on pressure and volume of the system as :

$$U + 34 + 3PV \text{ [All parameters in S.I. units]}$$

If state of fluid system changes from (100 Pa,  $3\text{m}^3$ ) to (400Pa,  $6\text{m}^3$ ) then incorrect option is :

- (A)  $\Delta U = 6300 \text{ J}$   
 (B)  $\Delta H = 8400 \text{ J}$   
 (C)  $W = -750 \text{ J}$   
 (D) Heat gained (q) = 5550 J

### SECTION-II : (Maximum Marks: 20)

This section contains 10 questions Candidates have to attempt any 5 questions out of 10. If more than 5 questions are attempted, then only first 5 attempted questions will be evaluated.

The answer to each question is an Integer Value Type questions.

For each question, enter the correct integer value (In case of non-integer value, the answer should be rounded off to the nearest Integer).

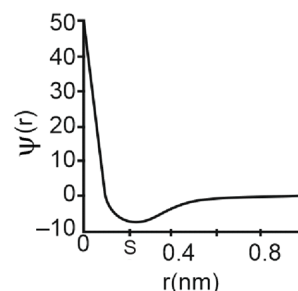
Answer to each question will be evaluated according to the following marking scheme:

*Full Marks* : +4 If correct answer is selected.

*Zero Marks* : 0 If none of the option is selected.

*Negative Marks* : -1 If wrong option is selected.

1. Consider the following plot of an atomic orbital associated with Hydrogen atom :

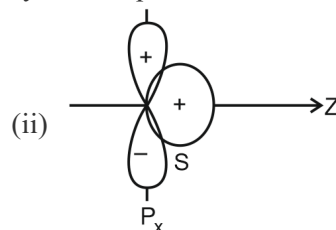


The plot does not touch point s.

The number of radial nodes with this spherically symmetric orbital is \_\_\_\_\_.

2. How many statements about chemical bonds are correct?

(i) During the formation of  $\text{H}_2$  molecule, a stage is reached where the net force of attraction balances the force of repulsion and system acquires maximum energy.



The given overlap is classified as a positive overlap between s and p atomic orbitals.

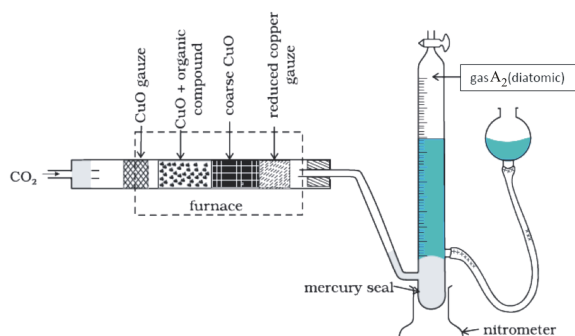
(iii) In case of  $\text{NH}_3$  and  $\text{H}_2\text{O}$  molecules, the HNH and HOH angles should be  $90^\circ$ .

(iv) In  $\text{PCl}_5$  the axial bond pairs suffer more repulsive interaction from the equatorial bond pairs,

(v) While an electron in an atomic orbital is influenced by one nucleus, in a molecular orbital it is influenced by two or more nuclei depending upon the number of atoms in the molecule.

(vi) The sigma ( $\sigma$ ) molecular orbitals are symmetrical around the bond-axis while pi ( $\pi$ ) molecular orbitals are not symmetrical.

3. The first ionization enthalpy values (in  $\text{kJ mol}^{-1}$ ) of group 13 elements are :  
579; 558; 589; 801; 577  
These are not in any particular order.  
Which of these belongs to Boron?
4. In how many of the following reaction, hydrogen peroxide acts as an oxidizing agent?  
(i)  $2\text{MnO}_4^- + 3\text{H}_2\text{O}_2 \rightarrow$   
(ii)  $2\text{MnO}_4^- + 6\text{H}^+ + 5\text{H}_2\text{O}_2 \rightarrow$   
(iii)  $\text{HOCl} + \text{H}_2\text{O}_2 \rightarrow$   
(iv)  $2\text{Fe}^{2+}(\text{aq}) + 2\text{H}^+(\text{aq}) + \text{H}_2\text{O}_2(\text{aq}) \rightarrow$   
(v)  $\text{I}_2 + \text{H}_2\text{O}_2 + 2\text{OH}^- \rightarrow$   
(vi)  $\text{PbS}(\text{s}) + 4\text{H}_2\text{O}_2(\text{aq}) \rightarrow$
5. In human body  $\text{H}_2\text{CO}_3/\text{HCO}_3^-$  buffer system helps to maintain pH of blood between p.26 and p.42, where p is \_\_\_\_\_.
6. The following experimental set-up is used for a popular method of estimation of an Element A :



In this method, the organic compound yields gas  $\text{A}_2$  on heating it with copper(II) oxide in the presence of  $\text{CO}_2$  gas. The mixture of gases is collected over potassium hydroxide solution in which  $\text{CO}_2$  is absorbed and volume of gas  $\text{A}_2$  is determined. The molar mass of gas  $\text{A}_2$  is \_\_\_\_\_  $\text{g mol}^{-1}$

7. How many the following sols are positively charged?  
(i) congo red sols  
(ii) Haemoglobin  
(iii) Sol of starch  
(iv) gold sol  
(v) eosin  
(vi) gelatin sol  
(vii) silver sol  
(viii)  $\text{CdS}$  sol  
(ix)  $\text{Sb}_2\text{S}_3$  Sol
8. How many of the following statements are incorrect?  
(i) Copper is used in several alloys e.g., brass (copper with tin)  
(ii) Nickel steel is used for cutting tools & crushing machines  
(iii) Zinc is used as a constituent of many alloys, e.g., german silver  
(iv) Cast iron is used for casting railway sleepers  
(v) Wrought iron is used in making agricultural implements  
(vi) Zinc is used for galvanising iron  
(vii) Stainless steel is used for making utensils
9. In Seliwanoff's test, D-Xylulose dehydrates very rapidly under acidic tests to give Furfural, which reacts with Resorcinol to give a coloured product. The number of carbon atoms in Furfural is \_\_\_\_\_
10. How many of the following cations give yellow sulphides on passing  $\text{H}_2\text{S}$  gas in the presence of dil.  $\text{HCl}$ ?  
 $\text{NH}_4^+$ ,  $\text{Pb}^{2+}$ ,  $\text{Mg}^{2+}$ ,  $\text{Al}^{3+}$ ,  $\text{Zn}^{2+}$ ,  $\text{Co}^{2+}$ ,  $\text{As}^{3+}$ ,  $\text{Ni}^{2+}$ ,  $\text{Ba}^{2+}$ ,  $\text{Mn}^{2+}$

SECTION-I : (Maximum Marks: 80)

This section contains **20 questions**. Each question has 4 options for correct answer. Multiple-Choice Questions (MCQs) **Only one option is correct**. For each question, marks will be awarded as follows:

*Full Marks* : +4 If correct answer is selected.

*Zero Marks* : 0 If none of the option is selected.

*Negative Marks* : -1 If wrong option is selected.

1. For two sets A and B,  $n(A) = 100$ ,  $n(B) = 120$ ,  
 $n(A \cap B) = 30$ ,  
 then  $n((A \times B) \cup (B \times A)) = \dots\dots$

- (A) 900  
 (B) 20100  
 (C) 23100  
 (D) 24000

2. If in a triangle ABC,  
 $\vec{BC} = \frac{\vec{u}}{|\vec{u}|} - \frac{\vec{v}}{|\vec{v}|}$  and  $\vec{AC} = \frac{2\vec{u}}{|\vec{u}|}$   
 where  $|\vec{u}| \neq |\vec{v}|$ , then value of

$\cos 2A + \cos 2B + \cos 2C$  is

- (A) 1  
 (B) 0  
 (C) -1  
 (D) -2

3. Which of the following can be the remainder when n is divided by 8, where n satisfies the equation

$$\cos\left(\frac{\pi}{4}(n^2 + 2n)\right) = \sin\left(\frac{\pi}{4}(n^2 + n + 1)\right),$$

(where n is an integer) is

- (A) 1  
 (B) 7  
 (C) 4  
 (D) 0

4. A monument ABCD stands on a level ground. At a point P on the ground the portions AB, AC, AD subtends angles  $\alpha$ ,  $\beta$ ,  $\gamma$  respectively. If  $AB = a$ ,  $AC = b$ ,  $AD = c$ ,  $AP = x$  and  $\alpha + \beta + \gamma = 180^\circ$  then  $x^2 =$

- (A)  $\frac{a}{a+b+c}$   
 (B)  $\frac{b}{a+b+c}$   
 (C)  $\frac{c}{a+b+c}$   
 (D)  $\frac{abc}{a+b+c}$

5. If a right angled triangle is inscribed in a rectangular hyperbola  $xy = 4$ , then the angle between the tangent at the right angled vertex and the hypotenuse of the triangle is

- (A)  $45^\circ$   
 (B)  $30^\circ$   
 (C)  $60^\circ$   
 (D)  $90^\circ$

6. If  $1, \alpha_1, \alpha_2, \dots, \alpha_{3n}$  be the roots of equation  $x^{3n+1} - 1 = 0$ . Also  $\omega, \omega^2$ , are imaginary cube roots of unity, then 
$$\frac{(\omega^2 - \alpha_1)(\omega^2 - \alpha_2) \dots (\omega^2 - \alpha_{3n})}{(\omega - \alpha_1)(\omega - \alpha_2) \dots (\omega - \alpha_{3n})} =$$
- (A)  $\omega$
- (B)  $-\omega$
- (C) 1
- (D)  $\omega^2$

7. Given that  $f(x)$  is a real valued non-constant differentiable function such that  $f(x) \cdot f'(x) \leq 0$ , for all real  $x$ , then it follows that:

- (A)  $f^2(x)$  is increasing function
- (B)  $f^2(x)$  is decreasing function
- (C)  $f(x)$  has no critical point
- (D)  $f(x) = 0$  have real root

8. Number of divisors of  $\sum_{r=0}^6 (-1)^r {}^{16}C_r$

- (A) 6
- (B) 10
- (C) 20
- (D) 16

9. Relation

$\left\{ \left[ (a' \wedge b')' \vee c \right] \wedge (a \vee c) \right\}'$   $a, b \in B$ , where  $B$  is Boolean algebra is equivalent to

- (A)  $(a' \wedge c')'$
- (B)  $(a' \wedge c')$
- (C)  $(a' \vee c')$
- (D) none of these

10.  $P$  is the parabola passing through the points  $(4, 4), (6, 8), (2, 4)$  and  $(0, 8)$ . Which of the following is false?

- (A) Vertex of  $P$  is  $\left(3, \frac{7}{2}\right)$
- (B) Length of Latus-Rectum of  $P$  is 2
- (C) equation of directrix is  $y - 3 = 0$
- (D) None of these

11. If for  $n \geq 3$ ,  $A^n = A^{n-2} + A^2 - I$ ,

where  $A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$ , then  $A^{49}$  is

- (A)  $\begin{bmatrix} 1 & 0 & 0 \\ 25 & 0 & 1 \\ 24 & 1 & 0 \end{bmatrix}$  (B)  $\begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 1 \\ 24 & 1 & 0 \end{bmatrix}$
- (C)  $\begin{bmatrix} 1 & 0 & 0 \\ 24 & 0 & 1 \\ 25 & 1 & 0 \end{bmatrix}$  (D)  $\begin{bmatrix} 1 & 0 & 0 \\ 24 & 0 & 24 \\ 24 & 1 & 0 \end{bmatrix}$

12. A circle  $x^2 + y^2 + 4x - 2\sqrt{2}y + c = 0$  is the director circle of circle  $S_1$  and  $S_1$  is the director circle of circle  $S_2$  and so on. If the sum of radii of all these circles is 2, then the value of  $c$  is
- (A) 1  
(B)  $3\sqrt{2}$   
(C)  $\sqrt{2}$   
(D)  $4\sqrt{2}$
13. If  $\phi_1(x) = x^2 + (k - 29)x - k$  and  $\phi_2(x) = 2x^2 + (2k - 43)x + k$ , both are factors of a cubic polynomial  $p(x)$  with leading coefficient 2, then value of  $p(0)$  can be
- (A) 75  
(B) -150  
(C) -75  
(D) 10
14. If  $\int \frac{6(x^{13} - x^8 + x^5) dx}{(3x^8 + 4x^6 + 2x^3 - 1)^2} = \frac{-x^6}{ax^8 + bx^6 + cx^3 - 1} + k$  where  $k$  is constant of integration, then  $(a + b + c)$  is equal to
- (A) 12  
(B) 13  
(C) 15  
(D) 9
15. The set of values of  $p$  for which  $f(x) = p^2 x - \int 2^{4-x^2} dx$  is increasing for all  $x \in \mathbb{R}$ , is
- (A)  $[-4, 4]$   
(B)  $(-\infty, -16] \cup [16, \infty)$   
(C)  $(-\infty, -4] \cup [4, \infty)$   
(D)  $[-16, 16]$
16. The degree of the differential equation whose general solution is given by  $y = (c_1 + c_2) \cos(x + c_3) - c_4 e^{x+c_5}$  where  $c_1, c_2, c_3, c_4, c_5$  are arbitrary constants, is-
- (A) 5  
(B) 4  
(C) 1  
(D) 2
17. The area bounded by  $y = \cos^{-1}(\sin x)$  and  $y = \sin^{-1}(\sin x)$  on the interval  $[0, \pi]$  is
- (A)  $\frac{\pi^2}{8}$   
(B)  $\frac{\pi^2}{6}$   
(C)  $\frac{\pi^2}{4}$   
(D) none of these

18. Equation of the acute angle bisector of the angle between the lines,

$$\frac{x-1}{1} = \frac{y-2}{1} = \frac{z-3}{1} \text{ and}$$

$$\frac{x-1}{1} = \frac{y-2}{1} = \frac{z-3}{-1} \text{ is:}$$

(A)  $\frac{x-1}{2} = \frac{y-2}{2}; z-3=0$

(B)  $\frac{x-1}{2} = \frac{y-2}{2} = \frac{z-3}{3}$

(C)  $x-1=0; \frac{y-2}{2} = \frac{z-3}{3}$

(D)  $\frac{x-1}{2} = \frac{y-2}{3}; z-3=0$

19. If

$$2f(x) + f(-x) = \frac{1}{x} \sin\left(\frac{x^2-1}{x}\right) \quad \forall x \in \mathbb{R} - \{0\}$$

then the value of  $\int_{\frac{1}{e}}^e f(x) dx$  is

(A) 0

(B) 1

(C) 2

(D) 3

20. Let  $t_n$  be the  $n^{\text{th}}$  term of the sequence.

$$\left(\frac{2^2}{1^2} - \frac{2}{1}\right)^{-1}, \left(\frac{3^3}{2^3} - \frac{3}{2}\right)^{-2}, \left(\frac{4^4}{3^4} - \frac{4}{3}\right)^{-3}, \dots$$

then  $\lim_{n \rightarrow \infty} (t_n)^{1/n}$  is equal to

(A)  $\frac{1}{e-1}$

(B)  $e-1$

(C) 1

(D) Doesn't exist

## SECTION-II : (Maximum Marks: 20)

This section contains 10 questions Candidates have to attempt any 5 questions out of 10. If more than 5 questions are attempted, then only first 5 attempted questions will be evaluated.

The answer to each question is an Integer Value Type questions.

For each question, enter the correct integer value (In case of non-integer value, the answer should be rounded off to the nearest Integer).

Answer to each question will be evaluated according to the following marking scheme:

**Full Marks** : +4 If correct answer is selected.

**Zero Marks** : 0 If none of the option is selected.

**Negative Marks** : -1 If wrong option is selected.

1. If  $\sqrt{2} \cos A = \cos B + \cos^3 B$ ,

$$\sqrt{2} \sin A = \sin B - \sin^3 B,$$

the find value of  $\left| 3 \sin(A-B) \right|$  is.....

2. Value of is

$$\left| \lim_{x \rightarrow \frac{3\pi}{4}} \frac{4\sin^2 x \cos x - \cos x + \sin x}{\sin x + \cos x} \right|$$

3. If lines given by the equation  $(3 \sec \theta + 5 \operatorname{cosec} \theta)x + (7 \sec \theta - 3 \operatorname{cosec} \theta)y + 11(\sec \theta - \operatorname{cosec} \theta) = 0$  pass through a fixed point P for all possible values of  $\theta$  (wherever defined), and if the maximum value of difference of distances of P and B (3, 4) from a point on the line  $x - y + 3 = 0$  is  $\lambda$  then find  $[\lambda]$  (where  $[\cdot]$  represent greatest integer function).

4. Two players A and B play a match which consists of a series of games (independent). Whoever first wins two games, not necessarily consecutive, wins the match. The probabilities of A's winning, drawing or losing a game against B are  $\frac{1}{2}$ ,  $\frac{1}{3}$ ,  $\frac{1}{6}$  respectively. If it is known that A won the match at the end of 11<sup>th</sup> game, If the probability that B win only one game is  $\frac{p}{q}$  (p and q are co-prime) then value of  $q - p$  is.....
5. Find the value of  $3b^2$  for which the common tangents to the ellipse  $\frac{x^2}{4} + \frac{y^2}{b^2} = 1$  and the parabola  $y^2 = 4x$  and their chord of contact form an equilateral triangle.
6. Find the number of 6 digit numbers which can be formed using the digits 1, 2, 3, 4, 5, 6 such that sum of the digits at even places is divisible by 3 (repetition of digits is not allowed).
7. If the minimum value of the expression  $|z| + |z - 1| + |z - 1 - i| + |z - i|$ , where  $z$  is a complex number and  $\sqrt{-1}$  is  $\lambda$ , then find  $\lambda^2$ .

8.  $\lim_{n \rightarrow \infty} \frac{\frac{1}{1.3.5.7} + \frac{1}{3.5.7.9} + \frac{1}{5.7.9.11} + \dots \text{upto } n \text{ terms}}{\frac{1}{3.7.11.15} + \frac{1}{7.11.15.19} + \frac{1}{11.15.19.23} + \dots \text{upto } n \text{ terms}}$   
is  $k$  then  $\frac{5}{22}k$  is equal to
9. A trapezium ABCD is inscribed into a semicircle of radius  $l$  so that the base AD of the trapezium is diameter and the vertices B and C lie on the circumference. Find the base angle  $\theta$  (in degree) of the trapezium ABCD which has the greatest perimeter.
10. If  $\sum_{i=1}^{18} (x_i - 8) = 9$  and  $\sum_{i=1}^{18} (x_i - 8)^2 = 45$ ,  
if the standard deviation of  $x_1, x_2, x_3, \dots, x_{18}$  is  $\frac{p}{q}$   
(p, q are coprime) then find  $p + q$ .

## Space for Rough Work



